Shivam Kumar Verma

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Experience

poiro.com **Jul 2026 – Present**
Full Stack Developer Intern *Python, FastAPI, MongoDB, Redis, Celery, Next.js, TypeScript, Cloudflare R2*

- **Creative Asset Manager:** Built one library where every upload and AI generation is saved through a single path, so users can organise assets into folders to build a full film or video. Stored each generation's model, prompt and reference images, and added Elements: reusable character, prop and location image sets with @handles for the prompt box.
- **Async Celery Worker Runtime:** Gave each Celery worker one long-lived event loop and a shared Mongo client and Redis pool, cutting connection churn and ending dead-connection write failures after database restarts.
- **MCP Agent Access:** Built the auth layer that lets AI assistants like Claude run Poiro apps for a user, capped by scopes and daily and per-run credit limits; RS256 tokens are verified via JWKS, refresh tokens rotate, and grants are checked on every request so revoking access is instant.

klimb.io **Nov 2025 – May 2026**
Full Stack Developer Intern *TypeScript, Express.js, PostgreSQL, Redis, AWS*

- **Usage-Based Billing Ledger:** Built a credit ledger to replace on/off feature flags, metering every use of 6+ AI features per customer and debiting it from their balance, which let klimb charge customers by usage.

Projects

CodeSM | GitHub | Live | *Code Platform, Node.js, Express.js, BullMQ, Redis, PostgreSQL/Drizzle, Docker, AWS S3*

- **Sandboxed Code Execution:** Ran each submission in an isolated Docker container with no network, a read-only filesystem and CPU, memory and time limits, containing infinite loops, fork bombs and network calls.
- **Async Grading Queue:** Moved grading to BullMQ workers so the API returns instantly; each submission runs all its test cases in one reused container and stops at the first failure.

DeepDoc | GitHub | *RAG, Next.js, TypeScript, Azure OpenAI, Pinecone, PostgreSQL/Drizzle, Vercel Blob*

- **RAG Pipeline:** Built a PDF Q&A pipeline that splits documents into section-aware chunks, searches only the active document's vectors, and rewrites follow-up questions from chat history before retrieval.
- **Retrieval Ranking and Tuning:** Ranked retrieved chunks by score, filtered out weak and duplicate matches, and packed the best into a fixed token budget; tuned chunking and retrieval settings across 100+ configurations.

Yuvro | GitHub | Live | *Replit (Web IDE), Python, FastAPI, PostgreSQL, Kubernetes, AWS S3, WebSockets*

- **Content-Addressed Storage:** Stored every file once by its content hash, deduplicating identical files across projects, with a scheduled garbage collector that deletes unreferenced files.
- **Per-Project Containers on Kubernetes:** Ran each project in its own on-demand Kubernetes container with a live WebSocket terminal, file editor, app preview and database browser, keeping user code off the main servers.

Skills

Languages: Python, TypeScript, SQL
Backend: FastAPI, Node.js, Express.js, Celery, BullMQ, asyncio, REST APIs, WebSockets, JWT/OAuth
Databases: PostgreSQL, MongoDB, Redis, Pinecone
Cloud & DevOps: Docker, Kubernetes, AWS S3, Cloudflare R2
AI: RAG, Vector Search, Azure OpenAI, MCP
Frontend & Tools: React, Next.js, Git, Turborepo

Education

Indian Institute of Information Technology Nagpur **Nov 2022 – May 2026**
B.Tech in Computer Science and Engineering *Nagpur, Maharashtra*

Achievements

- **LeetCode:** Top 5% globally with a **1800**+ contest rating and **800**+ problems solved.
- **CodeChef:** **3-star** coder with a 1700+ rating.